

Exercises :

L.1.5.

Saturation : $I : J^\infty$ is the ideal-theoretic analogue to $\overline{V(I) \setminus V(J)}^Z$

$$\Rightarrow V(I : J^\infty) = \overline{V(I) \setminus V(J)}$$

$\Rightarrow$ removing the part of $V(I)$ that is contained in $V(J)$

① Matrices

```
A1 = matrix(ZZ, [
    [0, -1, 1, 0, 0],
    [0, 0, 0, -1, 1]
])

A2 = matrix(ZZ, [
    [0, 1, 0, 1, 2],
    [0, 0, 1, 1, 0]
])

A1hat = matrix(ZZ, [
    [1, 1, 1, 1, 1],
    [0, -1, 1, 0, 0],
    [0, 0, 0, -1, 1]
])

A2hat = matrix(ZZ, [
    [1, 1, 1, 1, 1],
    [0, 1, 0, 1, 2],
    [0, 0, 1, 1, 0]
])

Cay = matrix(ZZ, [
    [0, -1, 1, 0, 0, 0, 0, 1, 0, 1, 2],
    [0, 0, 0, -1, 1, 0, 0, 0, 1, 1, 0],
    [1, 1, 1, 1, 1, 0, 0, 0, 0, 0, 0],
    [0, 0, 0, 0, 0, 1, 1, 1, 1, 1, 1]
]);
```

✓ 0.0s

② Rings and ideals

```
# -----  
# Target rings  
# -----  
  
Rx, x = polynomial_ring(QQ, ["x1", "x2", "x3", "x4", "x5"])  
Ry, y = polynomial_ring(QQ, ["y1", "y2", "y3", "y4", "y5"])  
  
# -----  
# Raw toric ideals from OSCAR  
# -----  
  
I_A1hat_raw = toric_ideal(transpose(A1hat))  
I_A2hat_raw = toric_ideal(transpose(A2hat))  
  
# -----  
# Move them into the rings Rx and Ry  
# -----  
  
S1 = base_ring(I_A1hat_raw)  
S2 = base_ring(I_A2hat_raw)  
  
phi1 = hom(S1, Rx, x)  
phi2 = hom(S2, Ry, y)  
  
I_A1hat = ideal(Rx, [phi1(f) for f in gens(I_A1hat_raw)])  
I_A2hat = ideal(Ry, [phi2(f) for f in gens(I_A2hat_raw)])  
  
println("I_A1hat in QQ[x1,...,x5] =")  
println(I_A1hat)  
println()  
  
println("Generators of I_A1hat:")  
for f in gens(I_A1hat)  
    println(" ", f)  
end  
println()  
  
println("I_A2hat in QQ[y1,...,y5] =")  
println(I_A2hat)  
println()  
  
println("Generators of I_A2hat:")  
for f in gens(I_A2hat)  
    println(" ", f)  
end  
println()
```

Output:

```
I_A1hat in QQ[x1,...,x5] =  
Ideal (x2*x3 - x4*x5, x1^2 - x2*x3)  
  
Generators of I_A1hat:  
x2*x3 - x4*x5  
x1^2 - x2*x3  
  
I_A2hat in QQ[y1,...,y5] =  
Ideal (y2*y4 - y3*y5, -y1*y4 + y2*y3, -y1*y5 + y2^2, y1*y4^2 - y3^2*y5)  
  
Generators of I_A2hat:  
y2*y4 - y3*y5  
-y1*y4 + y2*y3  
-y1*y5 + y2^2  
y1*y4^2 - y3^2*y5
```

③

Cayley
config

```
R, v = polynomial_ring(QQ, [
    "x1", "x2", "x3", "x4", "x5",
    "y1", "y2", "y3", "y4", "y5"
])

x1, x2, x3, x4, x5, y1, y2, y3, y4, y5 = v

Cay = matrix(ZZ, [
    0 -1 1 0 0 0 0 1 0 1 2;
    0 0 0 -1 1 0 0 1 1 0;
    1 1 1 1 1 0 0 0 0 0;
    0 0 0 0 0 1 1 1 1 1
])

I_Cay_raw = toric_ideal(transpose(Cay))
S = base_ring(I_Cay_raw)

phiC = hom(S, R, v)
I_Cay = ideal(R, [phiC(f) for f in gens(I_Cay_raw)])
```

Output :

```
I_Cay in QQ[x1,...,x5,y1,...,y5] =
Ideal with 21 generators

Generators of I_Cay:
y2*y4 - y3*y5
-x2*y5 + x4*y4
x3*y4 - x5*y5
-y1*y4 + y2*y3
-x1*y4 + x3*y3
x1*y3 - x2*y4
-y1*y5 + y2^2
-x1*y4 + x5*y2
-x1*y5 + x3*y2
x2*y2 - x4*y3
x1*y2 - x2*y5
-x1*y3 + x5*y1
-x1*y2 + x3*y1
x1*y1 - x2*y2
x2*x3 - x4*x5
x1^2 - x2*x3
y1*y4^2 - y3^2*y5
x1*y4^2 - x5*y3*y5
-x2*y1*y4 + x4*y3^2
-x1*x2*y4 + x4*x5*y3
x2*y4^3 - x5*y3^2*y5
```

min generators

4

Saturation

```
Jy = ideal(R, [y1,y2,y3,y4,y5])
Jx = ideal(R, [x1,x2,x3,x4,x5])

I_from_Cay_to_x = eliminate(saturation(I_Cay, Jy), [y1,y2,y3,y4,y5])
I_from_Cay_to_y = eliminate(saturation(I_Cay, Jx), [x1,x2,x3,x4,x5])

println("After saturating by <y1,...,y5> and eliminating y:")
for f in gens(I_from_Cay_to_x)
|   println(" ", f)
end
println()

println("After saturating by <x1,...,x5> and eliminating x:")
for f in gens(I_from_Cay_to_y)
|   println(" ", f)
end
println()
```

✓ 0.5s

After saturating by <y1,...,y5> and eliminating y:

$x_2^2 x_3 - x_4^2 x_5$

$x_1^2 - x_4^2 x_5$

After saturating by <x1,...,x5> and eliminating x:

$y_2^2 y_4 - y_3^2 y_5$

$-y_1^2 y_4 + y_2^2 y_3$

$-y_1^2 y_5 + y_2^2$

$y_1^2 y_4^2 - y_3^2 y_5$

I_{A_1}, I_{A_2}

4.1.7

For an affine variety V_A we know that the variety is invariant under transforms that leave the row span of A invariant (Thm. 1.3.14.)

The reason is that the toric ideal is defined via $\ker(A)$. But $\text{row } A \perp \ker(A)$, so $\ker(A)$ remains invariant.

We need to show that

$$\text{row}(\text{Cay}(A_\bullet)) = \text{row}(\text{Cay}(T \cdot A_\bullet + u_\bullet))$$

Two matrices A, B have the same row span if $M \cdot A = B$.

Here choose

$$M = \begin{pmatrix} T & \begin{pmatrix} m_1 & \dots & m_r \\ \vdots & & \vdots \\ u_{nd} & \dots & u_{nd} \end{pmatrix} \\ 0 & \underline{1} \end{pmatrix}$$

$$\text{s.t.} \quad M \cdot \text{Cay}(A_\bullet) = \text{Cay}(T \cdot A_\bullet + u_\bullet)$$

4.1.9

$$\dim X_{A_0} = \text{rank}(\mathbb{Z}A_0)$$

We know that $\dim X_{A_0} = \text{rank}(\text{Cay}(A_0)) - \ell$

Take $\text{Cay}(A_0)$ and perform invertible column operations

$$a_j^{(i)} \rightarrow a_j^{(i)} - a_1^{(i)} \quad j = 2, \dots, n_i$$

Then the new column set of $\text{Cay}(A_0)$ is

$$\underbrace{\left\{ \begin{pmatrix} a_1^{(1)} \\ \vdots \\ e_1 \end{pmatrix}, \dots, \begin{pmatrix} a_1^{(e)} \\ \vdots \\ e_e \end{pmatrix} \right\}}_{\text{lin. independent}} \sqcup \underbrace{\left\{ \begin{pmatrix} a_j^{(i)} - a_1^{(i)} \\ \vdots \\ 0 \end{pmatrix} \mid 1 \leq i \leq \ell, 2 \leq j \leq n_i \right\}}$$

lin. independent

Recall:

$$\begin{aligned} \mathbb{Z}A_0 &= \overline{\sum_j \overline{\sum_i \mathbb{Z}(a_j^{(i)} - a_1^{(i)})}} \\ &= \text{span}_{\mathbb{Z}} \left(\begin{pmatrix} a_1^{(1)} - a_1^{(1)} \\ \vdots \\ a_1^{(e)} - a_1^{(e)} \\ 0 \end{pmatrix} \right) \end{aligned}$$

So:

$$\begin{aligned} \dim(\text{col}(\widetilde{\text{Cay}(A_0)})) &= \ell + \dim(\mathbb{Z}A_0) \\ &= \ell + \text{rank}(\mathbb{Z}(A_0)) \end{aligned}$$

$$\Rightarrow \dim(X_{A_0}) = \text{rank}(\text{Cay}(A_0)) - \ell = \text{rank}(\mathbb{Z}(A_0))$$

$$\dim(\operatorname{col}(\operatorname{Cay}(A_{\bullet})) = \operatorname{rank}(\operatorname{Cay}(A_{\bullet}))$$

$$\begin{aligned}\dim(X_{A_{\bullet}}) &= \operatorname{rank}(\operatorname{Cay}(A_{\bullet})) - \ell \\ &= \operatorname{rank}(\mathbb{Z}(A_{\bullet}))\end{aligned}$$